

Detecting departure and destination aerodromes from ADS-B

Version 3 — what the residual errors are made of

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Summary

Version 3 follows a review of [version 2](#). Auditing V2 against its own outputs found four review points only partly delivered and one delivered with a defect. Fixing them changed the results and, more usefully, changed what the *remaining* errors are made of.

Three things came out of it.

The out-of-area rule was pre-empting good answers. The border test won unconditionally over a qualifying aerodrome match, so Ponta Delgada — which sits about 8 NM inside the western edge of the ingestion box — had its departures labelled “out of area” with the aircraft still on the runway. A qualifying aerodrome now wins and the border label is the fallback.

Six of the seven most frequent departure errors were the same mistake: a scheduled civil airport losing to an adjacent military airfield. Izmir to Çiğli Airbase, Naples to Grazzanise, Cagliari to Decimomannu, Catania to Sigonella, Gdańsk to Cewice — 7 to 25 NM apart, and in every case the truth had scheduled service and the prediction did not. Ranking candidates on a distance penalised for aerodromes without scheduled service removes that class of error entirely.

Out-of-area is reported separately now, and it does not work as well as V2 implied. Precision is high — when the rule says out-of-area it is right about 93% of the time — but recall is only 10 to 15%. The concept is right and the geometric test is weak, for a reason given in [Section 1](#).

Table 1: Departure detection: the production algorithm against endpoint abstention, headline and with out-of-area stripped out.

Sample	Method	ADEP cov	ADEP acc	in-area cov	in-area acc
2024-06	M0_current	60.66%	97.55%	65.83%	97.64%
2024-06	M7_abstain	72.41%	98.21%	77.70%	98.35%
2025-06	M0_current	63.59%	97.45%	69.29%	97.53%
2025-06	M7_abstain	74.85%	98.24%	80.05%	98.50%

What changed since version 2

! Important

- **Out-of-area precedence fixed** — an aerodrome match now beats the border label.
- **Scheduled-service tie-break added** — candidates ranked on distance penalised for aerodromes without scheduled service, swept rather than assumed.
- **Out-of-area reported explicitly** — recall, precision and in-area figures beside the headline, so naming skill and abstention skill cannot be conflated.
- **Per-side thresholds actually run** — and found not to help ([Section 4](#)).
- **Residual errors analysed** rather than described ([Section 3](#)).

Version 1 and version 2 remain published unchanged. Their numbers are not restated here and are not comparable to these, because the scoring changed.

1 Reporting out-of-area

A flight from Dubai to Amsterdam has a departure aerodrome that no European ADS-B feed can name: the aircraft entered the area already airborne. V2 made 00A a valid answer and a valid truth, which was right. What it did not do was report the two abilities separately, and they are different abilities — naming an aerodrome, and recognising that no aerodrome can be named.

That matters because roughly one departure in twelve is genuinely out of area, so a method could buy accuracy by labelling liberally and an undecomposed table would show only the improvement.

Table 2: Out-of-area detection for departures. Only M7 and M8 emit the label; the other methods never do, so their recall is zero by construction.

Sample	Method	truly OOA	labelled OOA	OOA recall	OOA precision
2024-06	M7_abstain	7.94%	0.83%	9.69%	92.25%
2024-06	M8_extrapolate	7.94%	1.67%	20.36%	96.78%
2025-06	M7_abstain	8.30%	1.31%	14.84%	93.91%
2025-06	M8_extrapolate	8.30%	1.17%	13.05%	92.79%

High precision, low recall. The rule fires when a track *endpoint* sits within 30 NM of the bounding box edge — but an inbound flight’s first recorded sample appears wherever a receiver happens to pick it up, which is usually well inside the boundary rather than at it. Most out-of-area departures therefore never look like border cases at all.

The concept is sound; the geometric test is the wrong instrument. A phase-based test would fit the physics better: a track whose first sample is **already in the cruise, neither near an aerodrome nor descending toward one**, has an origin outside the observed area regardless of where it is. That is not implemented here and is the clearest single improvement available (Section 9).

2 The military-airfield confusion

The most frequent departure errors in version 2 formed one pattern:

Table 3: In every case the truth had scheduled service and the prediction did not.

Truth (scheduled)	Predicted (not scheduled)	Apart
LTBJ Izmir	LTBL Çiğli Airbase	14.9 NM
LIRN Naples	LIRM Grazzanise Air Base	14.1 NM
LIEE Cagliari	LIED Decimomannu Air Base	7.2 NM
LICC Catania	LICZ Sigonella Navy Air Base	7.9 NM
EPGD Gdańsk	EPCE Cewice Naval Air Base	24.6 NM

A departing aircraft’s first recorded fix is often already a few miles into the climb, and where a military field sits between the airport and the departure track, the nearest-aerodrome rule takes it. Aviation knowledge resolves this without any new data: a commercial flight almost certainly used the aerodrome with scheduled service.

Candidates are therefore ranked on **effective distance** — real distance plus a fixed penalty for aerodromes without scheduled service — so a military field has to be clearly nearest rather than merely nearest. A penalty of zero reproduces the unbiased ranking exactly, which is why it is swept.

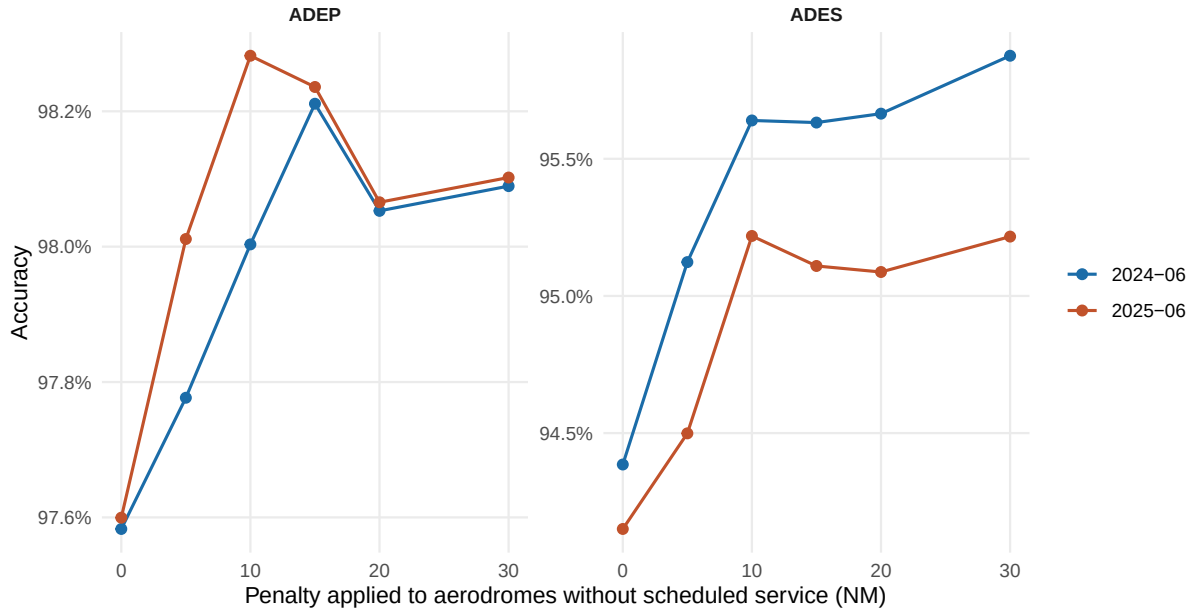


Figure 1: Accuracy against the scheduled-service penalty, both samples and both sides. Coverage moves by less than 0.6 pp across the whole range, so this is very nearly a free choice; the curve peaking and turning is what shows the penalty is correcting a bias rather than simply favouring larger aerodromes.

Accuracy peaks between 10 and 15 NM on both sides in both samples, then turns over. Coverage falls by under 0.6 pp across the entire range. **10 NM is the defensible choice** — it is at or within a whisker of the peak in all four curves, and the flat top means the exact value is not load-bearing.

3 Where the remaining errors are

The tie-break does not reduce errors uniformly. It removes one class and introduces another, and the honest way to report that is by composition.

Table 4: Composition of the 783 most frequent departure errors after the tie-break.

Class	Flights	Share
Scheduled → unscheduled (the class the tie-break targets)	0	0.00%
Unscheduled → scheduled (the tie-break's own cost)	464	59.26%
Truth out-of-area, an aerodrome named	192	24.52%
Truth an aerodrome, labelled out-of-area	23	2.94%

The class the tie-break targets is now **empty**. What dominates instead is its mirror image: genuine movements at non-scheduled aerodromes — Biggin Hill, Northolt — pulled to the

nearby hub. That is the price of the preference, it is visible, and it is smaller than what it replaced, which is why accuracy rose.

Table 5: Most frequent departure confusions after the fixes.

Sample	Truth	Predicted	Flights	Apart
2024-06	LCLK	OLBA	83	112.0 NM
2025-06	EGKB	EGLC	59	10.5 NM
2024-06	EGKB	EGLC	57	10.5 NM
2025-06	OOA	LPPD	55	—
2024-06	OOA	LPPD	54	—
2025-06	LFMD	LFMN	49	13.1 NM
2025-06	OOA	GMMN	35	—
2024-06	OJAI	OLBA	32	128.5 NM
2025-06	OOA	UGTB	30	—
2024-06	EGWU	EGLL	26	5.2 NM
2025-06	EGWU	EGLL	26	5.2 NM
2024-06	LFMD	LFMN	24	13.1 NM

Two further groups are visible and neither is addressable by threshold tuning. **Long-range misassignments** — Larnaca predicted as Beirut, 112 NM apart — are tracks whose first sample is far from either, where nearest-aerodrome has no real information. **Dense terminal areas** — Biggin Hill against London City at 10.5 NM, Cannes against Nice at 13.1 NM — need runway-level discrimination, which the HexAero layouts OPDI already computes could provide and this study does not use.

4 Per-side thresholds do not help

Version 2 noted that departures and arrivals behave differently and suggested fitting their thresholds separately. Doing it properly: fit each side’s threshold on **2024-06** by taking the most coverage available subject to beating the production algorithm’s accuracy, then apply to **2025-06**.

Both sides select the **same** cell, 40 NM and 15,000 ft AGL. The frontier’s shape is similar enough on the two sides that no separation emerges, so the asymmetry between departures and arrivals is not something a threshold can fix — it is in how much information the endpoint carries, not in where the cut-off should sit.

Table 6: Validation on 2025-06 of thresholds fitted on the other sample.

Configuration	ADEP cov	ADEP acc	ADES cov	ADES acc
Production (M0)	63.59%	97.45%	61.97%	95.55%
M7 default (20 NM / 15,000 ft)	74.85%	98.24%	63.76%	95.11%
M7 fitted on 2024-06 (40 NM / 15,000 ft)	75.90%	97.86%	66.63%	93.75%

The fitted point transfers **for departures** — more coverage at better accuracy than production, on a sample it was not fitted to. It does not transfer for arrivals: it buys coverage at a real cost in accuracy. This is the third independent time this study has found that arrivals resist what works for departures.

5 Everything else, unchanged

The cascade still collapses onto its fallback rung, the aircraft denylist still outperforms the allowlist it replaced, trajectory extrapolation still fails on arrivals, cleaning is still incompatible with endpoint-based arrival detection, and `track_id` still merges and fragments flights at the rates version 2 measured. Those results are unaffected by the changes here and are not repeated; see [version 2](#).

Table 7: All methods, both June samples, at the default configuration. In-area figures are in the summary table for M0 and M7; they track the headline closely for every method.

Sample	Method	ADEP cov	ADEP acc	ADES cov	ADES acc
2024	M8 extrapolate	76.29%	95.53%	53.20%	67.07%
2024	M6 cascade	85.41%	84.50%	85.39%	75.42%
2024	M1 traffic	83.42%	86.51%	81.96%	78.55%
2024	M7 abstain	72.41%	98.21%	64.41%	95.63%
2024	M3 vert_rate	63.79%	97.08%	59.97%	95.01%
2024	M5 min_alt	79.26%	76.92%	79.05%	72.40%
2024	M0 current	60.66%	97.55%	59.45%	94.40%
2024	M2 on_ground	27.33%	97.70%	30.34%	96.41%
2025	M8 extrapolate	77.41%	95.60%	51.73%	65.15%
2025	M1 traffic	84.90%	86.97%	83.45%	78.24%
2025	M6 cascade	86.95%	84.92%	86.86%	75.19%
2025	M7 abstain	74.85%	98.24%	63.76%	95.11%
2025	M3 vert_rate	66.72%	96.90%	62.54%	95.53%
2025	M5 min_alt	81.59%	78.17%	81.30%	73.41%
2025	M0 current	63.59%	97.45%	61.97%	95.55%
2025	M2 on_ground	28.87%	95.43%	33.38%	92.74%

6 How it works in code

Everything below is in `benchmarks/adep_ades.py` and is pure Spark column expressions and window functions — no `applyInPandas`, no UDFs, matching the convention of the pipeline it is meant to inform. This section is the description a reader needs to check the results or reimplement them.

6.1 Building tracks

State vectors are filtered to aeroplanes and then grouped into tracks by OPDI's own `TrackProcessor._add_track_id`, imported and called rather than reimplemented — `track_id` continuity with every published release depends on it. Tracks are materialised to S3 once per sample and every method reads them back, so all methods see byte-identical input.

The aircraft filter is a **denylist evaluated on a left join**:

```
out = sv.join(db, F.lower(sv.icao24) == F.col("_ic"), "left")
is_rotorcraft = F.col("ac_class").isin(["H", "G", "T"])
out.filter(~F.coalesce(is_rotorcraft, F.lit(False)) & ...)
```

The `coalesce` is the whole point. An airframe absent from the aircraft database gets `ac_class = NULL`, so `is_rotorcraft` is `NULL`; without the `coalesce` the predicate would be `NULL` and the row would be dropped. Unknown must mean keep, because 34% of state vectors come from airframes the database does not contain.

6.2 Finding candidate aerodromes

A track-by-aerodrome cross join is intractable, so candidates are bucketed on integer degree cells first. Each aerodrome is replicated into its 3×3 cell neighbourhood at load time, so an equi-join on `(cell_lat, cell_lon)` cannot miss a match that sits just across a cell boundary, and exact great-circle distance is computed only on the survivors:

```
apt.crossJoin(offss)                # offs = 9 (dy, dx) pairs
  .withColumn("cell_lat", F.floor("apt_lat").cast("int") + F.col("dy"))
```

One degree of latitude is 60 NM, so a 3×3 neighbourhood covers any radius up to about 40 NM — which bounds the usable distance threshold and is why the sweep stops there.

6.3 Endpoints

The first and last sample of each track are picked with two row numbers over the same window, forward and reversed, rather than a self-join:

```
w = Window.partitionBy("track_id").orderBy("event_time")
sv.withColumn("_rn", F.row_number().over(w))
  .withColumn("_rr", F.row_number().over(w.orderBy(F.col("event_time").desc()))))
  .filter((F.col("_rn") == 1) | (F.col("_rr") == 1))
  .withColumn("role", F.when(F.col("_rn") == 1, "adept").otherwise("ades"))
```

The join to aerodromes is a **LEFT** join. An endpoint over open sea has no aerodrome in its cell neighbourhood, and an inner join would discard exactly the endpoints that carry the out-of-area signal before it can be read.

6.4 Ranking candidates

Candidates are ordered by *effective* distance — real distance plus a fixed penalty for aerodromes without scheduled service (Section 2):

```
penalty = F.when(F.col("apt_scheduled") == "yes", F.lit(0.0)) \
  .otherwise(F.lit(sched_penalty_nm))
eff_nm  = F.col("dist_nm") + penalty
best    = Window.partitionBy("track_id", "role").orderBy(F.col("eff_nm").asc_nulls_last())
```

`asc_nulls_last` matters: an endpoint with no candidate at all still produces a row, with `NULL` distance, which must sort last rather than first. M1 passes `sched_penalty_nm=0` so it stays a faithful recreation of `traffic`'s rule.

6.5 The abstention test

Per side, an aerodrome is emitted only when the endpoint is credibly at it:

```
ok = (F.col(f"{side}_dist_nm") <= d) & (  
  F.col(f"{side}_on_ground")  
  | (F.col(f"{side}_elev_known") & (F.col(f"{side}_agl_ft") <= h))  
)
```

Three details carry weight. `on_ground` satisfies the height test on its own, because a missing barometric altitude on the surface is normal. `elev_known` guards the height comparison, because OurAirports has no elevation for 3.09% of large and medium aerodromes and treating a missing value as sea level is wrong by the entire threshold at a high field. And height is **above field elevation**, not above the ellipsoid:

```
F.col("baro_altitude") * 3.28084 - F.coalesce(F.col("apt_elev_ft"), F.lit(0.0))
```

6.6 Out-of-area

The border test scales the longitude margin by latitude, because a degree of longitude shrinks toward the pole and an unscaled margin would be twice as strict in northern Norway as in the Canaries:

```
dlat = margin_nm / 60.0  
dlon = dlat / F.greatest(F.cos(F.radians(lat)), F.lit(0.1))
```

Precedence is the fix described in the summary — the aerodrome match is tried first and the border label is the fallback:

```
F.when(ok, F.col(side)).otherwise(  
  F.when(F.col(f"{side}_at_border"), F.lit(00A)))
```

The border branch deliberately carries **no** height condition. A flight leaving the observed area is at cruise level by definition, so any height cut-off would reject precisely the endpoints the branch exists to catch.

Ground truth is relabelled symmetrically: an aerodrome outside the bounding box, or an ICAO code that does not resolve in OurAirports, becomes 00A, so the prediction and the truth live in the same alphabet.

6.7 Extrapolation (M8)

The projection is a constant-rate continuation over the final five minutes, capped at 60 NM. Time-to-ground is signed so that arrivals descending forward and departures climbing backward both yield a positive number:

```
signed_vs = F.when(F.col("role") == "ades", -F.col("vs_ms")).otherwise(F.col("vs_ms"))  
t_ground = F.when(signed_vs > 0.5, F.col("alt_m") / signed_vs)  
reach_nm = F.least(t_ground * F.col("gs_ms") * 0.000539957, F.lit(max_project_nm))
```


The > 0.5 gate means a track ending in level flight produces no projection at all rather than a division by something near zero. Departures project backwards by adding 180° to the track angle.

6.8 Scoring

Ground truth is the denominator throughout, so a flight never seen counts as a miss rather than an absent row:

```
p = ident.join(pred, "track_id", "left")
j = gt.join(p, ["icao24", "callsign", "day"], "left")
```

2.42% of (icao24, callsign, date) keys collide — same-day multi-leg rotations — and are resolved by taking the track whose start is nearest the actual off-block time, rather than leaving it to whichever row the join happened to produce:

```
w = Window.partitionBy("icao24", "callsign", "day", "gt_adept", "gt_ades") \
    .orderBy(F.abs(F.unix_timestamp("t_start")
        - F.unix_timestamp("gt_aobt")).asc_nulls_last())
```

The out-of-area decomposition then partitions ground truth on `gt_<side> == 00A` and reports recall and precision on that partition, and coverage and accuracy on its complement. `gt_ooa_share` comes out identical for every method scored against the same sample, which is the check that the decomposition is correct.

6.9 Cascade attribution

`diagnose_cascade` keeps every rung’s answer side by side rather than collapsing straight to a coalesce, records which rung supplied each answer, and scores that rung against a control **on the same flights**. That is what makes “this rung is 95% accurate” comparable with “the rung it displaced would have been 99% accurate here”, and it is the measurement that showed the vertical-trend methods contribute nothing.

7 Recommendations

7.1 Already implemented and validated in this study

These need no further research. They are in `benchmarks/adept_ades.py` and are measured above.

Change	Evidence
Treat out-of-area as an answer, not a silence	Section 1
Prefer a qualifying aerodrome over the border label	Section 1
Rank candidates with a scheduled-service penalty (10 NM)	Section 2

Change	Evidence
Drop only <i>confirmed</i> non-aeroplanes; keep unidentified airframes	V2
Restrict candidates to large and medium aerodromes	V2
Report naming skill and abstention skill separately	Section 1

7.2 Outstanding in the production pipeline

None of the above has been applied to `src/opdi/pipeline/flights.py`. The production flight list still uses the altitude-trend vote and still emits a null for both “outside the observed area” and “not determined”. A patch proposal follows so that the change can be reviewed rather than assumed.

8 Production patch proposal

Scope is `src/opdi/pipeline/flights.py`. `tracks.py` is untouched, so `track_id` continuity with every published release is unaffected — this changes the flight list, not track construction.

Replace `_categorize_landing_take_off`, which counts the sign of a 5-sample rolling-mean altitude difference and drops anything ambiguous, with the endpoint rule:

1. Take the first and last state vector of the track.
2. Find the nearest aerodrome by **effective distance** — great-circle distance plus 10 NM for aerodromes without scheduled service — among large and medium aerodromes only.
3. Emit that aerodrome when the endpoint is within 20 NM of it and no more than 15,000 ft above its elevation, or is flagged `on_ground`.
4. Otherwise, if the endpoint lies within 30 NM of the ingestion boundary, emit the out-of-area marker.
5. Otherwise emit null.

Add two columns to the flight list, `adep_source` and `ades_source`, taking `aerodrome`, `out_of_area` or `undetermined`. This is the change with the highest value per line: those three states are currently collapsed into a single null, and they mean entirely different things to a downstream user.

Version. New `version` string for the flight list. Existing event types keep theirs, so published v0.0.2 data stays reproducible.

Aerodrome elevation comes from OurAirports and is missing for 3.09% of large and medium aerodromes. Where it is missing, fall back to the `on_ground` flag rather than treating the elevation as sea level — at a 5,000 ft aerodrome that error is the entire height threshold.

Before applying, a reviewer should check: that arrivals are *not* switched to this rule on the strength of the departure result, since arrivals fail to replicate every time; that the aerodrome reference used in production carries `scheduled_service`; and that the out-of-area marker is agreed with downstream consumers before it appears in a published column.

9 Limitations

- **Out-of-area recall is 10–15%.** The geometric border test is the wrong instrument; a phase-based test — first sample already in the cruise, not near or descending toward an aerodrome — should be tried next, and is the single clearest improvement available.
- **The scheduled-service preference has a cost,** and it is now the largest single class of remaining departure errors: genuine movements at Biggin Hill and Northolt pulled to the nearby hub.
- **Dense terminal areas are unresolved.** Biggin Hill against London City, Cannes against Nice. Runway-level matching using the HexAero layouts OPDI already computes is the obvious next step and is not used here.
- **Six days,** two three-day samples a year apart.
- Everything carried over from version 2: surface vehicles cannot be excluded by declaration, `track_id` is taken as given, and cleaning is not applied.

10 Reproducing

```
python benchmarks/adeq_ades.py --days 2025-06-05 2025-06-06 2025-06-07 \  
    --months 202506 --methods all --sweep-abstain \  
    --error-pairs M7_abstain --per-airport M7_abstain \  
    --diagnose-cascade --control m1 --sched-penalty-nm 15 \  
    --results-dir <dir>
```

`--results-dir` writes the result tables as CSV to a local directory instead of S3. The bucket has a quota, and a write that exceeds it fails at the very end of a job — after the numbers are computed and printed but before they are stored.

Rendering this page needs no credentials, no database and no cluster.